Symmetry-Constrained Resonant Modes on H_4 Geometry

A Geometric Framework for Vibrational Field Dynamics

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Abstract

We present a geometric framework in which physical structure arises as stable resonant modes of a scalar field constrained by the symmetry of the Coxeter group H_4. The geometric scaffold is constructed from a hierarchy of φ-based symmetries—pentagonal (H_2), icosahedral (H_3), and four-dimensional polytope symmetry (H_4)—terminating at the 600-cell, a regular four-dimensional polytope with 120 vertices and 600 tetrahedral cells. A real scalar field is defined on the vertex graph of the 600-cell, and a prototype dynamical equation incorporating the H_4 graph Laplacian is introduced. The linear eigenmode spectrum of this system is computed explicitly and found to contain only nine distinct eigenvalues across 120 degrees of freedom, with a multiplicity pattern that includes the square sequence $(1, 4, 9, 16, 25, 36)$. Nonlinear extensions permit the formation of stable localised structures. The relationship between H_4 root systems and the E_8 lattice is discussed as a possible higher-dimensional embedding. This work establishes the mathematical and geometric foundations for Vibrational Field Dynamics (VFD), a resonance-based model of structure formation.

Keywords: Coxeter groups, H_4 symmetry, 600-cell, graph Laplacian, resonant eigenmodes, nonlinear lattice dynamics, E_8 lattice, structure formation.

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1 Introduction

Modern theoretical physics relies heavily on symmetry as a guiding principle for understanding the structure of matter and the behaviour of fields. From Noether's theorem linking symmetries to conservation laws [1] to the role of gauge symmetries in the Standard Model [2], physical laws often emerge from the mathematical properties of underlying symmetry groups. In most contemporary frameworks the starting point is a set of fields defined on spacetime, whose dynamics are constrained by internal symmetry groups.

An alternative perspective is to begin from geometry itself. Rather than treating symmetry as an auxiliary property of fields, one may ask whether the geometric structure of symmetry spaces can act as the *primary* organising principle from which physical structure emerges. In such an approach the geometry of symmetry manifolds constrains the allowable dynamical modes of fields defined upon them. Stable physical structures would then correspond to resonant solutions permitted by the geometry of the underlying symmetry space.

This idea has partial precedent. The role of geometry in general relativity [3], the geometric phase in quantum mechanics [4], and the extensive use of fibre bundles in gauge theory [5] all demonstrate that geometry is not merely a convenient language but an active ingredient in physical law. More recently, connections between polytope geometry and scattering amplitudes [6] have reinforced the view that geometric structures may encode physical information in ways that are not apparent from a Lagrangian-first perspective.

Vibrational Field Dynamics (VFD) explores this possibility from a specific geometric starting point. The central idea of VFD is that physical structures may be understood as stable resonant modes of a field constrained by a geometric symmetry hierarchy derived from φ-based polytope geometry. In particular, the theory focuses on the symmetry group H_4, whose geometric realisation is the *600-cell*—one of the regular four-dimensional polytopes. The 600-cell forms a closed symmetry shell constructed from tetrahedral cells and defined by a root system of 120 vectors. This structure represents the highest-dimensional regular polytope associated with the icosahedral symmetry family.

Within this framework the geometry of the 600-cell provides a natural boundary condition for the modes of a field defined on the associated symmetry manifold. Resonant solutions of such a field correspond to standing-wave configurations constrained by the symmetries of the polytope. These resonant eigenmodes form the candidate structures that may correspond to physical entities.

The purpose of this paper is to formalise the geometric scaffold underlying VFD. We begin by describing the symmetry hierarchy that leads from pentagonal geometry to the four-dimensional H_4 symmetry group (Section 2). We then describe the geometric role of the 600-cell as a symmetry boundary (Section 3), followed by the introduction of a field defined on the associated manifold and the derivation of a prototype dynamical equation (Section 4). Sections 5 and 6 address the resonant eigenmode spectrum and its interpretation as a mechanism for structure formation. Section 7 discusses the relationship between H_4 root systems and the E_8 lattice. Finally, Sections 8 and 9 outline implications and directions for future work.

The central hypothesis of VFD may be stated concisely: *physical structures correspond to stable resonant eigenmodes of a field constrained by discrete symmetry geometries derived from the H_4 root system.*

The objective of the present work is not to claim a complete physical theory but to establish the geometric and dynamical framework necessary for a resonance-based model

of structure formation. The purpose is to demonstrate that the H_4 geometry produces a concrete, computable, and highly structured eigenmode spectrum rather than to claim immediate correspondence with known particle physics.

2 The Symmetry Ladder of φ-Geometry

The geometric scaffold underlying VFD arises from a hierarchy of symmetries associated with the golden ratio $\varphi = (1 + \sqrt{5})/2 \approx 1.618$. The golden ratio appears naturally in geometries exhibiting pentagonal symmetry and plays a central role in the structure of the Platonic solids with icosahedral symmetry. This family of geometries forms a progression that extends across dimensions, which we term the *symmetry ladder*.

2.1 Pentagonal Symmetry and the Golden Ratio (H_2)

The starting point of the hierarchy is the regular pentagon. The pentagon possesses five-fold rotational symmetry described by the dihedral group D_5 of order 10 (five rotations and five reflections), which is the two-dimensional realisation of the Coxeter group H_2 [7]. The diagonals of the pentagon intersect in proportions determined by φ: the ratio of diagonal to side length equals the golden ratio. This relationship is not incidental—it is the defining geometric property that propagates throughout the higher-dimensional structures in the ladder.

The golden ratio satisfies the algebraic relation

$$\varphi^2 = \varphi + 1 \tag{1}$$

and is the positive root of the polynomial $x^2 - x - 1 = 0$. The number field $\mathbb{Q}(\sqrt{5})$ generated by φ is the minimal field extension of the rationals required to express the coordinates of vertices in the icosahedral symmetry family [8]. This algebraic fact underlies the structural unity of the entire symmetry ladder.

2.2 Icosahedral Symmetry (H_3)

In three dimensions the pentagonal symmetry extends to the pair of dual Platonic solids: the icosahedron and the dodecahedron. These solids share the same symmetry group, denoted H_3, which describes the full icosahedral symmetry group in three dimensions with order 120 [7].

The icosahedron has 12 vertices, 30 edges, and 20 triangular faces. The dodecahedron has 20 vertices, 30 edges, and 12 pentagonal faces. The two are dual structures: the vertices of one correspond to the face centres of the other. The geometry of both solids incorporates the golden ratio in their metric relationships. For example, the vertices of a regular icosahedron inscribed in a unit sphere may be expressed as permutations of [9]

$$(0,\ \pm 1,\ \pm\varphi). \tag{2}$$

The icosahedral group H_3 is generated by three reflections satisfying the Coxeter relations encoded in the Coxeter diagram for H_3, which features a bond of order 5 reflecting the underlying pentagonal geometry.

2.3 Four-Dimensional Symmetry (H_4)

The symmetry family continues into four dimensions through the Coxeter group H_4 [10]. This group describes the symmetry of two regular four-dimensional polytopes: the 600-cell and the 120-cell.

- The **600-cell** consists of 600 tetrahedral cells, 1200 triangular faces, 720 edges, and 120 vertices.
- The **120-cell** consists of 120 dodecahedral cells, 720 pentagonal faces, 1200 edges, and 600 vertices.

As in the three-dimensional case the two polytopes form a dual pair: the vertices of the 600-cell correspond to the cell centres of the 120-cell, and vice versa.

The Coxeter group H_4 has order 14 400 and is generated by four reflections. Its Coxeter diagram is

$$\underset{s_1}{\circ} - \underset{s_2}{\circ} - \underset{s_3}{\circ} \overset{5}{=} \underset{s_4}{\circ} \tag{3}$$

where the bond labelled 5 indicates the H-type (pentagonal) relationship that distinguishes this family from the classical A, B, D, and E families [10].

2.4 Termination of the Symmetry Ladder

Importantly, the H_n family terminates at H_4. There is no regular convex polytope in Euclidean space of dimension five or higher that possesses icosahedral-type symmetry [11]. The classification of regular polytopes shows that for $n \geq 5$ only three families exist: the simplex, the hypercube, and the cross-polytope, each with symmetry groups of type A and B respectively. The exceptional polytopes associated with H_3 and H_4 have no higher-dimensional analogues.

In this sense the 600-cell represents the *terminal member* of the non-crystallographic Coxeter family H_n within the family of regular polytopes.

Table 1: The symmetry ladder of φ-geometry.

Group	Dimension	Geometric Realisation	Order
H_2	2	Pentagon / Decagon	10
H_3	3	Icosahedron / Dodecahedron	120
H_4	4	600-cell / 120-cell	14 400

This ladder forms the geometric backbone of the VFD framework. Each step extends the symmetry structure into a higher dimension while preserving the underlying φ-based relationships. The termination at H_4 suggests that the 600-cell may play a distinguished role as a maximal symmetry boundary in a resonance-based model.

Among finite reflection groups, H_4 occupies a unique position: it is the highest-dimensional non-crystallographic Coxeter group associated with icosahedral symmetry. Unlike the infinite families A_n, B_n, and D_n, which extend to arbitrary dimension, the H_n family admits only three members. The resulting 600-cell therefore represents not merely a convenient choice of geometry but the maximal regular polytope available within the φ-based symmetry hierarchy. This is the principal reason H_4 is selected as the geometric substrate for VFD.

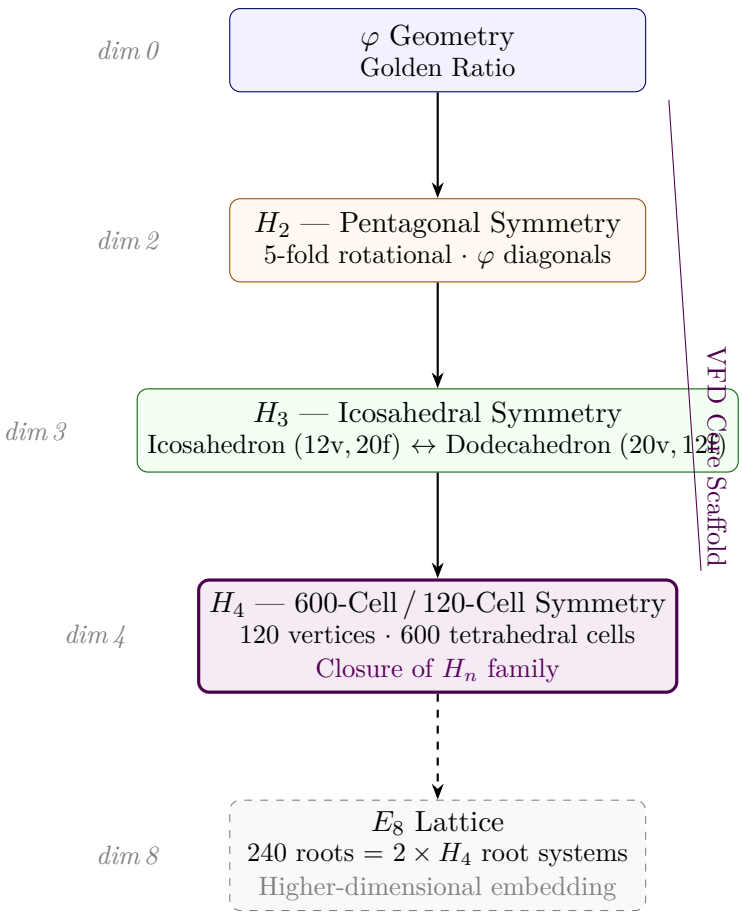


Figure 1: The symmetry ladder underlying Vibrational Field Dynamics. The hierarchy begins with φ-geometry and pentagonal symmetry (H_2), extends through the icosahedral group H_3, and closes at the H_4 symmetry of the 600-cell. The E_8 lattice (dashed) represents a possible higher-dimensional symmetry embedding containing H_4 root systems.

3 The 600-Cell as a Symmetry Boundary

The 600-cell occupies a distinguished position in the classification of regular polytopes. It is the four-dimensional polytope with the largest number of cells among the regular 4-polytopes and possesses the richest symmetry structure. In the VFD framework the 600-cell serves as the *maximal symmetry shell* within the H_n Coxeter family—the closed geometric constraint surface on which the dynamical field is defined.

3.1 Geometric Properties

The 600-cell is a convex regular 4-polytope whose boundary consists of 600 regular tetrahedra [12]. Its key combinatorial data are listed in Table 2.

Table 2: Combinatorial data for the 600-cell.

Element	Count
Vertices	120
Edges	720
Faces (triangles)	1200
Cells (tetrahedra)	600

Each vertex of the 600-cell is surrounded by 12 edges and 20 tetrahedral cells, giving it a local structure related to the icosahedron. Indeed, the *vertex figure* of the 600-cell—the cross-section obtained by slicing near a vertex—is a regular icosahedron [12]. This means that the local neighbourhood of every vertex carries the full H_3 symmetry, providing a direct geometric link between the three-dimensional and four-dimensional levels of the symmetry ladder.

3.2 The H_4 Root System

The vertices of the 600-cell, when the polytope is centred at the origin and inscribed in a unit sphere in $\mathbb{R}^4$, form the root system of the Coxeter group H_4 [10, 13]. This root system consists of 120 vectors.

Using the quaternion representation, the vertices of the 600-cell correspond to the 120 elements of the *binary icosahedral group* $2I$, a double cover of the icosahedral rotation group [14]. These 120 unit quaternions can be partitioned as:

- 24 vertices forming a 24-cell (corresponding to the binary tetrahedral group), and
- 96 vertices forming the remaining shell.

The coordinates of the 120 vertices can be written explicitly using φ and its conjugate $\hat{\varphi} = (1 - \sqrt{5})/2 = -1/\varphi$. Representative vertex coordinates include permutations and even sign changes of [15]:

$$\tfrac{1}{2}(\pm 1, \pm 1, \pm 1, \pm 1), \qquad (0, 0, 0, \pm 1), \qquad \tfrac{1}{2}(\pm\varphi, \pm 1, \pm\hat{\varphi}, 0). \tag{4}$$

3.3 Graph Structure

For the purposes of defining a discrete field theory, the relevant structure is the *graph* of the 600-cell. Two vertices are connected by an edge if and only if they are nearest neighbours in the polytope. Each vertex has exactly 12 nearest neighbours, giving the 600-cell graph a regular valency of 12.

The adjacency structure of this graph encodes the local symmetry constraints that any field defined on the vertices must respect. The 600-cell graph is *vertex-transitive* (all vertices are equivalent under the symmetry group) and *edge-transitive*, ensuring that the constraint structure is uniform across the entire shell.

3.4 Local Icosahedral Structure

An important structural feature of the 600-cell graph is that the *vertex figure* at each vertex is an icosahedron [12]. Equivalently, the 12 nearest neighbours of any vertex are arranged in the geometry of the icosahedral symmetry group H_3. This means that the prototype VFD dynamics combines two distinct layers of symmetry:

- **Global H_4 closure** at the level of the full 600-cell shell (120 vertices, $|H_4| = 14\,400$).
- **Local H_3 coupling** at the level of each vertex neighbourhood (12 neighbours arranged as an icosahedron).

For the graph Laplacian, the update rule at vertex i depends entirely on the neighbour set $N(i)$. Because those 12 neighbours carry icosahedral symmetry, every local coupling step is effectively an icosahedral averaging operation. The field does not oscillate on an arbitrary 12-regular graph; it oscillates on a graph whose local shell at every site is an icosahedron.

This two-layer structure—global H_4, local H_3—plausibly contributes to the strong compression and regularity of the Laplacian spectrum computed in Section 5.2. In particular, the appearance of the square-degeneracy sequence $(\ell+1)^2$ in the low-lying eigenvalues (Table 4) is consistent with the fact that the 600-cell vertices lie on S^3, while the icosahedral local geometry ensures that the discrete graph captures important features of the continuous symmetry at short range.

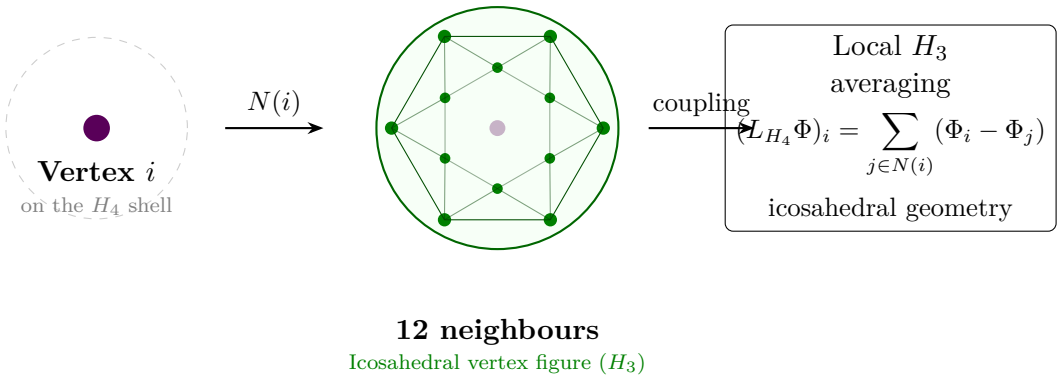


Figure 2: Local geometry of the 600-cell graph. Each vertex i has 12 nearest neighbours arranged as an icosahedron (the vertex figure). The graph Laplacian couples the field through repeated local H_3-symmetric averaging, embedded within the globally H_4-symmetric shell.

3.5 The 600-Cell as a Closed Constraint Shell

The combination of these properties—maximal symmetry within the H_n family, a well-defined root system, icosahedral vertex figures, and a regular graph structure—makes the 600-cell a natural candidate for a geometric constraint boundary. In the VFD framework we interpret the 600-cell as a *closed symmetry shell*: a maximal geometric structure that constrains the modes of a field defined on its vertices.

The maximality property is essential. Because H_4 is the terminal member of the non-crystallographic Coxeter family, the 600-cell is not a truncation or approximation of a higher-dimensional object within this family. It is the *final* regular structure, and its symmetries define the maximal set of geometric constraints available from the φ-based hierarchy.

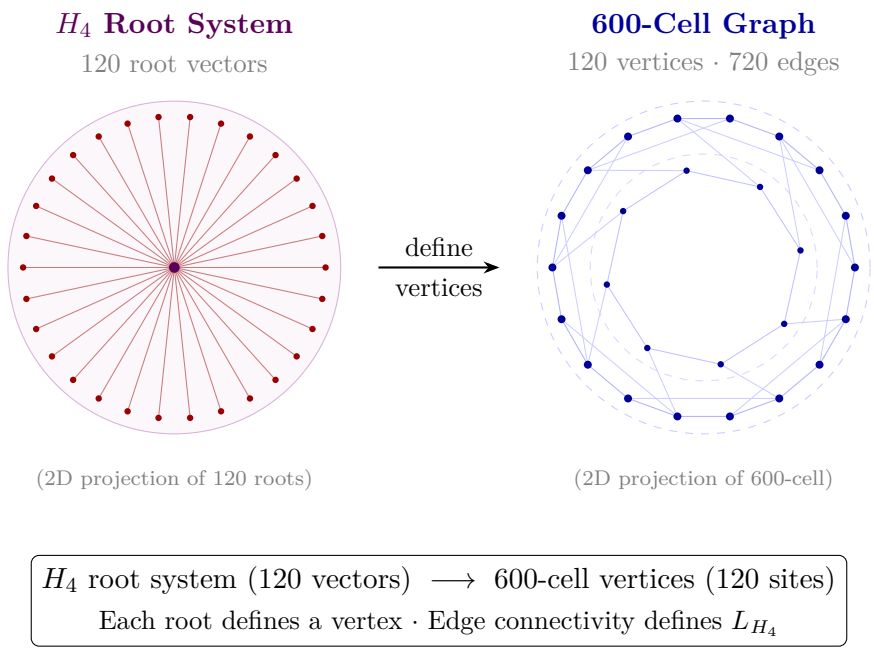


Figure 3: The H_4 root system (left) consists of 120 vectors whose endpoints define the vertices of the 600-cell (right). The edge connectivity of the 600-cell graph defines the adjacency structure used to construct the graph Laplacian L_{H_4}.

4 Field Definition on the H_4 Manifold

4.1 The Continuous Setting

In the most general formulation one considers a real scalar field defined on a manifold associated with the H_4 symmetry group. Let $\mathcal{M}$ denote the symmetry manifold—for instance, the orbit space of a point under the action of H_4, or a smooth manifold whose isometry group contains H_4. The field is a smooth function

$$\Phi : \mathcal{M} \times \mathbb{R} \longrightarrow \mathbb{R} \tag{5}$$

where the first argument represents position on the manifold and the second represents time. The dynamics of Φ are governed by a wave-type equation incorporating the geometric structure of $\mathcal{M}$ through its Laplace–Beltrami operator.

However, the continuous setting requires a precise specification of $\mathcal{M}$ as a Riemannian manifold, which introduces additional choices (metric, topology) beyond the discrete symmetry data. For the purposes of establishing the foundational framework we adopt a discrete approach that depends only on the combinatorial structure of the 600-cell.

4.2 The Discrete Field on the 600-Cell Graph

Let $V(H_4)$ denote the set of 120 vertices of the 600-cell. We define a real scalar field on this vertex set:

$$\Phi_i(t), \qquad i \in V(H_4), \quad |V(H_4)| = 120 \tag{6}$$

where i indexes the vertices and t denotes time. The field value $\Phi_i(t)$ represents the amplitude of the field at vertex i at time t. The state of the system at any time is therefore a vector in $\mathbb{R}^{120}$.

The space of all such field configurations inherits the symmetry of the 600-cell: the Coxeter group H_4 acts on $\mathbb{R}^{120}$ by permuting the vertex labels according to the symmetries of the polytope. This action decomposes the space of field configurations into irreducible representations of H_4, a fact that will be central to the analysis of eigenmodes.

4.3 The H_4 Graph Laplacian

To express the symmetry-constrained coupling between field values at neighbouring vertices, we introduce the graph Laplacian associated with the 600-cell connectivity structure.

> **Definition.** Let $G = (V, E)$ be the graph of the 600-cell, where $V = V(H_4)$ is the vertex set and E is the edge set. For a function $\Phi : V \to \mathbb{R}$, the *graph Laplacian* is defined by
>
> $$\boxed{(L_{H_4}\,\Phi)_i \;=\; \sum_{j\in N(i)} \Big(\Phi_i - \Phi_j\Big)} \tag{7}$$
>
> where $N(i)$ denotes the set of vertices adjacent to vertex i in the graph.

Since the 600-cell graph is 12-regular (every vertex has exactly 12 neighbours), the Laplacian can also be written in matrix form as

$$L_{H_4} = 12\,I - A \tag{8}$$

where A is the 120×120 adjacency matrix of the 600-cell graph and I is the identity matrix.

The graph Laplacian is a symmetric positive semi-definite matrix. Its null space is spanned by the constant vector (the trivial mode), and its nonzero eigenvalues encode the connectivity structure of the graph. Because the 600-cell graph is vertex-transitive under H_4, the eigenspaces of L_{H_4} are invariant subspaces of the permutation representation of H_4, ensuring that the eigenvalue decomposition respects the symmetry of the polytope.

4.4 Prototype VFD Field Equation

The simplest dynamical model consistent with these constraints is a nonlinear oscillator system defined on the graph. The prototype VFD field equation is

$$\boxed{\ddot{\Phi}_i + \omega_0^2\,\Phi_i + \lambda\,(L_{H_4}\,\Phi)_i + \beta\,\Phi_i^3 = 0} \tag{9}$$

where:

- ω_0 is a baseline oscillation frequency, providing a restoring force independent of geometry;
- λ is a coupling constant governing the strength of geometric interactions between neighbouring vertices;
- β is a nonlinear coefficient controlling the amplitude-dependent self-interaction of the field;
- the overdot denotes differentiation with respect to time.

This equation represents a system of 120 coupled nonlinear oscillators constrained by the symmetry geometry of the 600-cell. The Laplacian term ensures that the evolution of the field respects the adjacency relations defined by the H_4 structure, while the nonlinear term allows the formation of stable localised modes.

Table 3: Terms in the prototype VFD field equation.

Term	Physical Role
$\ddot{\Phi}_i$	Inertia / oscillatory dynamics
$\omega_0^2\,\Phi_i$	Baseline restoring force
$\lambda\,(L_{H_4}\,\Phi)_i$	Geometric coupling to neighbours
$\beta\,\Phi_i^3$	Nonlinear self-interaction

The equation is Hamiltonian, derivable from the energy functional

$$H = \sum_i \left[\frac{1}{2}\dot{\Phi}_i^2 + \frac{\omega_0^2}{2}\Phi_i^2 + \frac{\lambda}{4} \sum_{j\in N(i)} (\Phi_i - \Phi_j)^2 + \frac{\beta}{4}\Phi_i^4 \right]. \tag{10}$$

This ensures energy conservation and provides a variational foundation for the dynamics.

5 Resonant Eigenmodes

5.1 The Linear Eigenmode Problem

To understand the basic resonant structure of the system we first consider the linearised version obtained by setting $\beta = 0$:

$$\ddot{\Phi}_i + \omega_0^2\,\Phi_i + \lambda\,(L_{H_4}\,\Phi)_i = 0. \tag{11}$$

Seeking harmonic solutions of the form $\Phi_i(t) = A_i\,e^{i\Omega t}$, we obtain the matrix eigenvalue problem

$$\boxed{(\omega_0^2\,I + \lambda\,L_{H_4})\,A = \Omega^2\,A} \tag{12}$$

This is a standard symmetric eigenvalue problem for the 120×120 matrix $M = \omega_0^2 I + \lambda L_{H_4}$. Since L_{H_4} is positive semi-definite with eigenvalues $0 = \mu_1 \le \mu_2 \le \cdots \le \mu_{120}$, the resonance frequencies are

$$\Omega_k^2 = \omega_0^2 + \lambda\,\mu_k, \qquad k = 1, \ldots, 120. \tag{13}$$

The eigenvectors $A^{(k)}$ represent the symmetry-constrained normal modes of the field on the H_4 manifold. Each eigenmode corresponds to a standing-wave configuration defined over the vertices of the 600-cell.

5.2 First-Pass Spectrum of the 600-Cell Graph Laplacian

As a first explicit computation within the proposed H_4-constrained field model, we determine the spectrum of the 600-cell graph Laplacian $L_{H_4} = 12I - A$, where A is the adjacency matrix of the 600-cell graph. Using the standard 120-vertex realisation and nearest-neighbour edge connectivity, the graph is confirmed to be 12-regular with 720 edges.

The Laplacian spectrum contains only **nine distinct eigenvalues**, listed in ascending order with their multiplicities in Table 4.

Table 4: Eigenvalues and multiplicities of the 600-cell graph Laplacian $L_{H_4} = 12I - A$. The nine distinct eigenvalues compress 120 degrees of freedom into nine resonance sectors.

Sector k	Eigenvalue μ_k	Multiplicity d_k	Note
1	0	1	$= 1^2$
2	$9 - 3\sqrt{5} \approx 2.292$	4	$= 2^2$
3	$10 - 2\sqrt{5} \approx 5.528$	9	$= 3^2$
4	9	16	$= 4^2$
5	12	25	$= 5^2$
6	14	36	$= 6^2$
7	$10 + 2\sqrt{5} \approx 14.472$	9	
8	15	16	
9	$9 + 3\sqrt{5} \approx 15.708$	4	

$\sum d_k = 1 + 4 + 9 + 16 + 25 + 36 + 9 + 16 + 4 = 120.$ Eigenvalues involving $\sqrt{5}$ lie in $\mathbb{Q}(\sqrt{5}\,)$.

The spectrum was obtained by constructing the adjacency matrix of the 600-cell graph using the vertex coordinates and inner-product criterion described in Appendix A, then performing symmetric eigendecomposition of the Laplacian matrix $L_{H_4} = 12I - A$ using standard numerical linear algebra methods.

This result is noteworthy for two reasons. First, the 120-dimensional system collapses into only nine distinct resonance levels, indicating a highly compressed mode structure. Second, the first six multiplicities form the perfect-square sequence

$$d_1, \ldots, d_6 \;=\; 1^2,\ 2^2,\ 3^2,\ 4^2,\ 5^2,\ 6^2 \tag{14}$$

which is characteristic of the degeneracy pattern of harmonic analysis on the 3-sphere S^3, where the eigenspaces of the Laplace–Beltrami operator have dimension $(\ell + 1)^2$ for angular momentum quantum number ℓ. The fact that this pattern persists for the first six sectors of the discrete 600-cell graph suggests that the low-lying modes closely approximate the continuous hyperspherical harmonics, with deviations appearing only in the upper part of the spectrum.

The four irrational eigenvalues—$9 \pm 3\sqrt{5}$ and $10 \pm 2\sqrt{5}$—take values in the number field $\mathbb{Q}(\sqrt{5}\,)$, confirming the algebraic connection to the golden ratio that pervades the H_n symmetry family. The remaining five eigenvalues $(0, 9, 12, 14, 15)$ are rational integers. Note that the irrational eigenvalues appear in conjugate pairs, as expected from the Galois symmetry of $\mathbb{Q}(\sqrt{5}\,)/\mathbb{Q}$.

Within the present framework this spectrum constitutes the first concrete output of the VFD prototype model: a discrete hierarchy of symmetry-constrained standing-wave sectors on the 600-cell graph. While no direct physical identification is claimed at this stage, the existence of such a structured spectral organisation supports the hypothesis that the H_4 geometry provides a nontrivial and highly ordered constraint space for resonant structure formation.

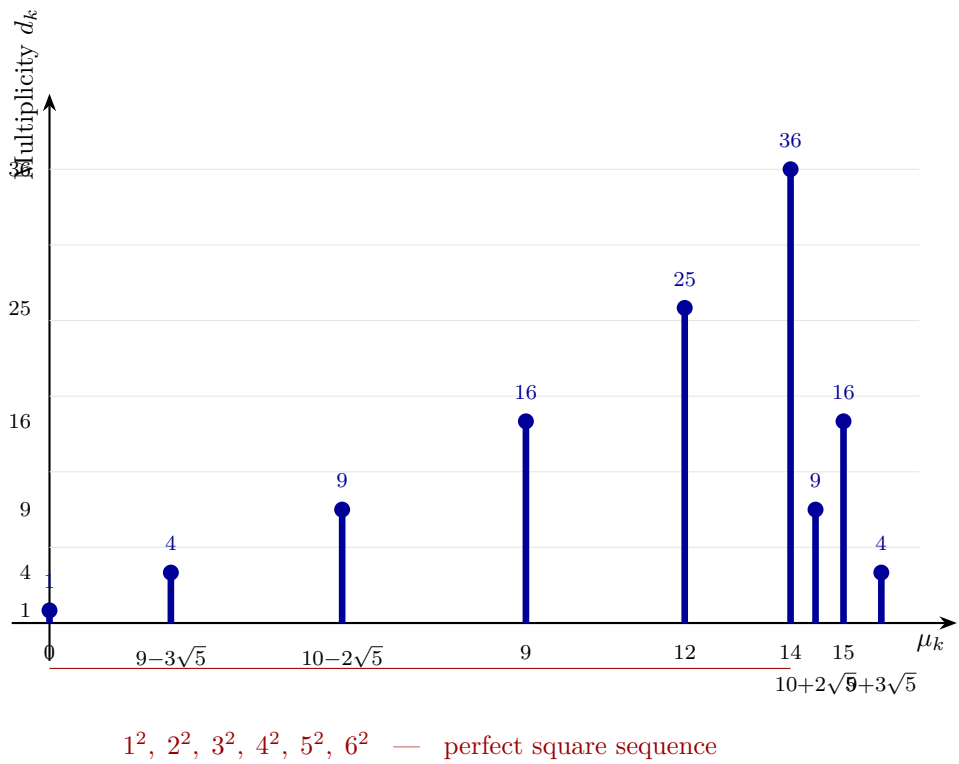


Figure 4: Distinct eigenvalues of the 600-cell graph Laplacian L_{H_4} and their multiplicities. The spectrum compresses 120 degrees of freedom into nine resonance sectors. The first six multiplicities (brace) follow the perfect-square sequence n^2, mirroring the degeneracy pattern of hyperspherical harmonics on S^3.

5.3 Symmetry and Degeneracy

The eigenvalue spectrum of L_{H_4} is highly structured due to the symmetry of the 600-cell graph. Because the graph is vertex-transitive under the action of H_4, the eigenspaces of the Laplacian decompose into irreducible representations of H_4 [16]. This means:

1. **Eigenvalues are highly degenerate.** The 120 modes do not yield 120 distinct frequencies; they organise into a much smaller number of distinct eigenvalues, each with a degeneracy equal to the dimension of the corresponding irreducible representation.
2. **The degeneracy pattern is determined by representation theory.** The specific multiplicities follow from the decomposition of the 120-dimensional permutation representation of H_4 into irreducible components.
3. **The lowest mode is always trivial.** The constant eigenvector (all components equal) corresponds to eigenvalue $\mu_1 = 0$ with degeneracy 1.

The spectral structure of the 600-cell graph Laplacian has been studied in the context of algebraic graph theory [17]. The eigenvalues reflect the algebraic structure associated with the φ-based symmetry family, with several taking values in the number field $\mathbb{Q}(\sqrt{5})$.

5.4 Mode Classification

The eigenmodes can be classified according to their transformation properties under H_4:

- **Trivial mode** (Ω_1): the uniform configuration where all vertices oscillate in phase. This is the lowest-frequency mode.
- **Vector modes** (low Ω): modes transforming as the fundamental 4-dimensional representation of H_4. These are analogous to dipole oscillations on a sphere.
- **Higher harmonics**: modes transforming as higher irreducible representations, with increasingly complex spatial patterns on the 600-cell graph.
- **Highest mode** ($\Omega_{\max}$): the mode with maximum variation between neighbouring vertices, corresponding to the largest Laplacian eigenvalue.

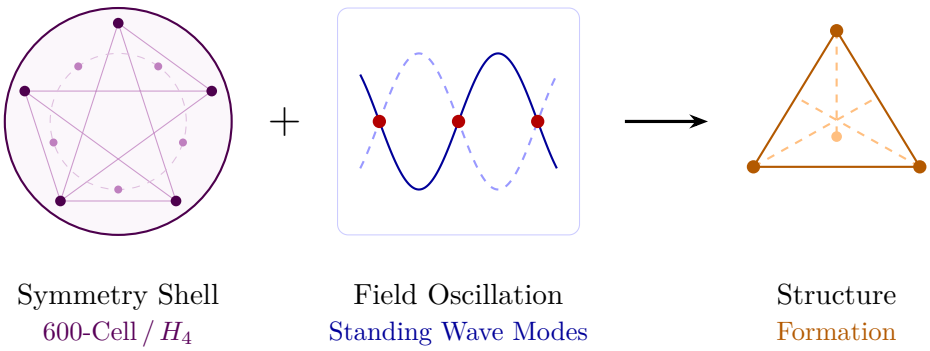
The progression from low to high modes represents increasing geometric complexity of the standing-wave pattern on the symmetry shell. In the VFD interpretation the low-lying modes with the simplest spatial structure are the primary candidates for stable physical configurations.

5.5 Comparison with Spherical Harmonics

The eigenmode decomposition on the 600-cell graph is the discrete analogue of the spherical harmonic decomposition of the Laplacian on S^2 or, more precisely, the harmonic analysis on the 3-sphere S^3 (since the vertices of the 600-cell lie on $S^3 \subset \mathbb{R}^4$).

On the continuous 3-sphere the eigenvalues of the Laplace–Beltrami operator are $\ell(\ell+2)$ with degeneracy $(\ell+1)^2$, and the eigenfunctions are the hyperspherical harmonics [18]. The 600-cell eigenmode spectrum can be understood as a *symmetry-reduced discretisation* of this continuous spectrum: the H_4 symmetry selects a finite subset of modes from the continuous family, and the discrete graph structure modifies the eigenvalues from their continuous values.

This relationship provides a bridge between the discrete VFD framework and continuous field theory on curved spaces.



Symmetry Constraints + Field Dynamics $\longrightarrow$ Resonant Eigenmodes $\longrightarrow$ Stable Structures

Figure 5: Resonant structure formation on the H_4 symmetry shell. A field defined on the H_4 symmetry manifold (left) forms standing-wave eigenmodes constrained by the geometry of the 600-cell (centre). Stable resonant modes correspond to structured configurations (right).

6 Structure Formation

The central claim of VFD is that stable physical structures correspond to persistent resonant configurations of the field on the symmetry shell. This section outlines the

mechanism by which the eigenmode spectrum derived in Section 5 may give rise to structure.

6.1 Linear Regime: Mode Selection

In the linear approximation ($\beta = 0$) the general solution is a superposition of eigenmodes:

$$\Phi_i(t) = \sum_{k=1}^{120} c_k \, A_i^{(k)} \cos(\Omega_k t + \phi_k) \tag{15}$$

where the coefficients c_k and phases ϕ_k are determined by initial conditions. In the absence of nonlinearity and dissipation all modes persist indefinitely and no particular configuration is selected.

However, even in the linear regime the symmetry structure imposes strong constraints. The degeneracy pattern ensures that only certain spatial configurations are consistent with the H_4 symmetry. Any physical mechanism that preferentially excites low-lying modes—for example, a cooling or energy-minimisation process—would produce configurations dominated by the lowest eigenmodes, which have the simplest and most symmetric spatial structure.

6.2 Nonlinear Regime: Mode Locking and Localisation

Restoring the nonlinear term $\beta \neq 0$ introduces qualitatively new phenomena.

Mode coupling. The cubic nonlinearity couples different eigenmodes. Energy initially concentrated in one mode can be redistributed among others, with the coupling strength depending on the overlap integrals of the eigenvectors. The H_4 symmetry constrains which modes can couple: the selection rules are determined by the Clebsch–Gordan decomposition of the tensor products of H_4 representations [19].

Amplitude-dependent frequencies. Nonlinearity causes the resonance frequency of each mode to depend on its amplitude:

$$\Omega_k^2(\text{eff}) \approx \omega_0^2 + \lambda\mu_k + \tfrac{3\beta}{2}|c_k|^2 + \cdots \tag{16}$$

This creates the possibility of *resonance locking*: when the effective frequencies of two or more modes become commensurate, energy transfer between them can stabilise, producing a persistent multi-mode configuration.

Localised modes. In nonlinear lattice systems it is well established that localisation can occur even in the absence of disorder. *Discrete breathers* (also known as intrinsic localised modes) are time-periodic, spatially localised solutions that exist generically in nonlinear lattice systems with sufficient anharmonicity [20, 21]. On the 600-cell graph the high connectivity (valency 12) and the rich symmetry structure provide a natural setting for such localised solutions.

6.3 Candidate Structures

Within the VFD framework, candidate physical structures are identified with specific types of persistent field configurations:

1. **Single-mode structures**: configurations dominated by a single eigenmode. These have the spatial symmetry of the corresponding irreducible representation of H_4 and oscillate at a well-defined frequency.
2. **Locked multi-mode structures**: configurations in which several modes are coupled by nonlinearity into a persistent bound state with a fixed amplitude and phase relationship.
3. **Localised breather-like structures**: spatially localised, time-periodic configurations arising from the nonlinear dynamics. These provide candidate particle-like excitations on the symmetry shell.

The stability of these configurations can in principle be analysed using standard methods: linear stability analysis around the candidate solution, Lyapunov exponent computation, and numerical integration of the full nonlinear system.

6.4 Geometric Packing of Modes

An additional structural feature arises from the geometry of the eigenmode spectrum. Because the eigenmodes are constrained by the H_4 symmetry, their spatial patterns on the 600-cell vertices are not arbitrary but must respect the polytope's geometric structure. Distinct stable modes occupy different geometric configurations on the same shell, analogous to electron orbitals on an atom but constrained by H_4 symmetry rather than SO(3).

The packing of multiple stable modes on the same symmetry shell—each occupying a distinct symmetry sector—provides a natural mechanism for generating a discrete spectrum of distinct structures from a single geometric substrate.

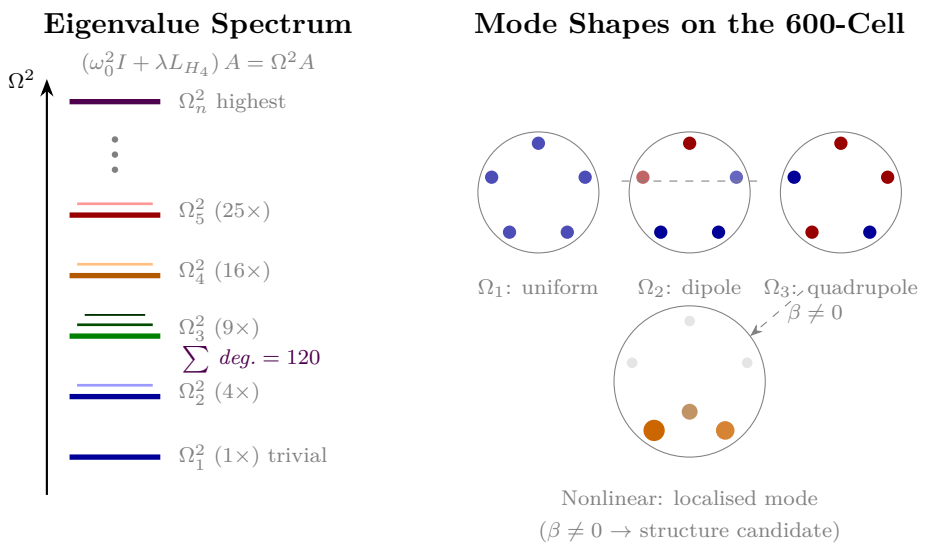


Figure 6: Schematic of the eigenmode spectrum on the 600-cell graph. *Left:* the eigenvalue spectrum Ω^2 is discrete and highly degenerate due to H_4 symmetry. *Right:* low-lying eigenmodes have simple spatial patterns (uniform, dipolar, quadrupolar); the nonlinear regime ($\beta \neq 0$) permits localised structures that are candidates for particle-like configurations.

7 Relationship to E_8 Symmetry

7.1 The E_8 Root System

The E_8 lattice is an eight-dimensional lattice with exceptional properties [22]. Its root system consists of 240 vectors of minimal length, forming the vertices of the Gosset polytope 4_{21}. The E_8 lattice is the densest sphere packing in eight dimensions [23] and plays a central role in string theory, where the gauge group $E_8 \times E_8$ appears in the heterotic string [24].

7.2 H_4 Inside E_8

A key mathematical relationship connects the H_4 and E_8 root systems. The E_8 root system admits a decomposition into two H_4 root systems under a specific embedding into $\mathbb{R}^8$ [25, 26]:

$$E_8 \text{ roots } (240) \supset H_4 \text{ roots } (120) \cup H_4' \text{ roots } (120). \tag{17}$$

This decomposition arises from a specific embedding of the H_4 root system into $\mathbb{R}^8$. The two H_4 copies are related by a rotation in the eight-dimensional space and together generate the full E_8 root system. Certain projections of the E_8 root system onto four-dimensional subspaces produce configurations with H_4 symmetry [25].

More precisely, there exists an orthogonal decomposition $\mathbb{R}^8 = V_1 \oplus V_2$ where each $V_i \cong \mathbb{R}^4$, such that the projection of the E_8 roots onto each four-dimensional subspace yields a copy of the H_4 root system. This is related to the Coxeter plane projection and the "golden ratio" embedding studied by several authors [26, 27].

7.3 Implications for VFD

Within the VFD framework the $E_8 \to H_4$ relationship suggests a possible hierarchy of models:

Level 1
: (this paper): A single H_4 system with 120 field sites on the 600-cell. This is the minimal model and is sufficient to establish the basic eigenmode spectrum and structure formation mechanism.

Level 2
: (future work): Two coupled H_4 systems, each carrying a 120-site field, embedded within an E_8 structure. The coupling between the two H_4 sectors would be governed by the eight-dimensional geometry and could give rise to additional structure: splitting of degenerate modes, inter-sector resonances, and new classes of coupled states.

Level 3
: (speculative): A field defined on the full E_8 root system, with the H_4 decomposition providing a natural basis for analysis.

It is important to emphasise that the E_8 connection is *not required* for the basic VFD framework presented in this paper. The single H_4 system on the 600-cell is self-contained and produces a well-defined eigenmode spectrum. The E_8 lattice is introduced here as a *possible higher-dimensional embedding* that may become relevant in future extensions of the theory, not as a necessary ingredient of the present model.

7.4 Relation to Physics

The appearance of E_8 in both string theory and the VFD symmetry ladder raises the question of whether a deeper connection exists. In heterotic string theory the $E_8 \times E_8$ gauge symmetry is a consequence of anomaly cancellation in ten-dimensional supergravity [24]. In the VFD context E_8 appears through the purely geometric route of polytope symmetry extension.

Whether these two appearances of E_8 are related—or whether the VFD geometric framework can make contact with established physical theories through the E_8 connection—remains an open question. We note the coincidence without claiming a specific relationship.

8 Implications

The framework developed in the preceding sections has several potential consequences, which we outline here as directions for investigation rather than as established results.

8.1 Discrete Structure Spectrum

The eigenmode spectrum of the H_4 graph Laplacian produces a finite, discrete set of allowed resonance frequencies and spatial configurations. If the VFD mechanism is physically realised, this discreteness provides a natural explanation for the observed discreteness of particle species: only those structures corresponding to stable eigenmodes (or nonlinear descendants thereof) would persist.

As computed in Section 5.2, the 600-cell graph Laplacian possesses exactly nine distinct eigenvalues with a highly organised multiplicity structure (Table 4). The compressed spectral hierarchy—nine resonance sectors spanning 120 degrees of freedom—suggests that the H_4 geometry naturally organises the field into a small number of collective mode families rather than an arbitrary collection of oscillatory states. This compression is a prerequisite for any resonance-based model aiming to reproduce the observed finiteness of the particle spectrum.

8.2 Stability Hierarchy

Not all eigenmodes are equally stable under perturbation. In the nonlinear regime some modes may be robust attractors while others are unstable saddle configurations. The stability hierarchy among the eigenmodes could provide an ordering of structures by persistence—potentially analogous to the observed hierarchy of particle lifetimes, where stable particles correspond to deeply stable modes and short-lived resonances correspond to marginally stable configurations.

8.3 Symmetry Constraints on Interactions

The H_4 symmetry imposes selection rules on mode coupling. Not all pairs of modes can exchange energy through the cubic nonlinearity; the coupling is governed by the representation-theoretic decomposition of tensor products. These selection rules constrain the allowed "interactions" between structure-modes in a manner reminiscent of

conservation laws in particle physics, though the specific content of these constraints has not yet been computed for the H_4 system.

8.4 Dimensional Considerations

The VFD field is defined on a four-dimensional symmetry manifold (the 600-cell in $\mathbb{R}^4$), while observable physics takes place in 3+1 spacetime dimensions. The relationship between the internal symmetry dimensions of the VFD framework and physical spacetime dimensions is not addressed in the present work and represents a significant open question. Possible approaches include interpreting the H_4 manifold as an internal symmetry space (analogous to Kaluza–Klein compactification) or as a constraint surface embedded in a higher-dimensional space.

9 Future Work

The present paper establishes the geometric and algebraic scaffold of VFD without performing explicit computations of the eigenmode spectrum or numerical simulations of the nonlinear dynamics. The following steps are necessary to develop the framework into a quantitative model.

9.1 Explicit Eigenmode Computation

The 120×120 graph Laplacian of the 600-cell should be explicitly constructed and diagonalised. This requires:

1. Generating the 120 vertex coordinates of the 600-cell.
2. Determining the adjacency matrix from the edge connectivity (nearest-neighbour criterion).
3. Computing $L_{H_4} = 12I - A$.
4. Diagonalising to obtain the eigenvalue spectrum $\{\mu_k\}$ and eigenvectors $\{A^{(k)}\}$.
5. Analysing the degeneracy pattern and identifying the corresponding H_4 representations.

This computation is straightforward numerically and would yield the first concrete VFD prediction: the resonance spectrum of the H_4 symmetry shell.

9.2 Numerical Simulation of the Nonlinear System

The full nonlinear field equation should be integrated numerically for various values of the parameters $(\omega_0, \lambda, \beta)$ and initial conditions. Key questions include:

- What stable localised modes (discrete breathers) exist on the 600-cell graph?
- How does the stability landscape depend on the nonlinearity parameter β?
- Are there attracting configurations that emerge from generic initial conditions?

Standard numerical methods for nonlinear lattice dynamics (symplectic integrators, Newton–Raphson continuation methods) are directly applicable [20].

9.3 Representation-Theoretic Analysis

A systematic decomposition of the 120-dimensional permutation representation of H_4 into irreducible components would provide the complete classification of eigenmodes and

their coupling rules. This analysis draws on the well-studied representation theory of finite reflection groups [28].

9.4 Mapping to Physical Observables

If the eigenmode spectrum can be computed, the most important next step is to determine whether any subset of the spectrum, or any ratio of eigenvalues, corresponds to known physical quantities—such as particle mass ratios or coupling constant relationships. Such a correspondence, even if approximate, would constitute significant evidence for the physical relevance of the framework.

9.5 Extension to E_8

As discussed in Section 7, the two-H_4 model embedded in E_8 represents a natural extension. This would double the number of field sites to 240 and introduce inter-sector coupling governed by the eight-dimensional geometry.

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A Numerical Construction of the 600-Cell Graph

This appendix describes the procedure used to construct the 600-cell graph and compute the Laplacian spectrum reported in Section 5.2. All steps are reproducible using standard numerical linear algebra software.

A.1 Vertex Coordinates

The 120 vertices of the 600-cell inscribed in the unit 3-sphere $S^3 \subset \mathbb{R}^4$ were generated using the quaternion representation. The vertex set consists of the 120 unit quaternions forming the binary icosahedral group $2I$ [14]. These can be constructed explicitly as the union of the following sets (all permutations of coordinates and even sign changes):

1. **8 vertices:** all permutations of $(\pm 1, 0, 0, 0)$.
2. **16 vertices:** $\frac{1}{2}(\pm 1, \pm 1, \pm 1, \pm 1)$ (all sign combinations).
3. **96 vertices:** all even permutations of $\frac{1}{2}(0, \pm 1, \pm\varphi, \pm\hat{\varphi})$, where $\varphi = \frac{1+\sqrt{5}}{2}$ and $\hat{\varphi} = \frac{1-\sqrt{5}}{2}$.

These yield $8 + 16 + 96 = 120$ unit vectors in $\mathbb{R}^4$.

A.2 Adjacency Matrix

Two vertices v_i and v_j are declared adjacent if and only if their Euclidean distance equals the edge length of the 600-cell inscribed in the unit sphere. For a unit-radius 600-cell the edge length is

$$d_{\text{edge}} = \frac{1}{\varphi} = \varphi - 1 = \frac{\sqrt{5}-1}{2} \approx 0.6180. \tag{18}$$

Equivalently, using the inner product, vertices v_i and v_j are adjacent if

$$\langle v_i, v_j \rangle = 1 - \frac{d_{\text{edge}}^2}{2} = 1 - \frac{3-\sqrt{5}}{4} = \frac{1+\sqrt{5}}{4} = \frac{\varphi}{2} \approx 0.80902. \tag{19}$$

In practice, adjacency is determined by checking whether $\left|\langle v_i, v_j \rangle - \varphi/2\right| < \varepsilon$ for a small numerical tolerance ε. The resulting adjacency matrix A is a 120×120 symmetric binary matrix with exactly 12 ones in each row, confirming the 12-regular structure. The total number of edges is $\frac{1}{2} \times 120 \times 12 = 720$, matching the known combinatorial data for the 600-cell.

A.3 Laplacian and Diagonalisation

The graph Laplacian is computed as $L_{H_4} = 12I - A$. Standard eigenvalue decomposition (e.g. `numpy.linalg.eigh` or equivalent) yields 120 eigenvalues. Grouping by numerical equality (within floating-point tolerance) produces the nine distinct eigenvalues and multiplicities reported in Table 4.

The closed-form expressions for the irrational eigenvalues ($9 \pm 3\sqrt{5}$ and $10 \pm 2\sqrt{5}$) were identified by comparing the numerical values against algebraic expressions in $\mathbb{Q}(\sqrt{5}\,)$ and verified to match within machine precision.

Code used to generate the 600-cell graph and compute the Laplacian spectrum will be released in a companion repository.